

A choice-free model for the dual mixed volume characterization

Solution packet for arXiv:1408.6796, Question 3.5

11 August 2026

Abstract

We answer Question 3.5 of Jiménez–Villanueva affirmatively in relative consistency form. In every model of $\mathbf{ZF} + \mathbf{DC}$ in which all sets of reals have the Baire property, separate radial additivity already forces the continuity missing from the source theorem. Consequently the asserted characterization by dual mixed volume holds in every dimension. Shelah’s 1984 consistency theorem supplies such a model from $\mathbf{Con}(\mathbf{ZF})$, so no inaccessible-cardinal assumption is required.

1 The question and the answer

Write $\mathcal{S}_0^n$ for the star bodies in $\mathbb{R}^n$, with radial addition, and recall

$$\tilde{V}(L_1, \dots, L_n) = \frac{1}{n} \int_{S^{n-1}} \rho_{L_1}(u) \cdots \rho_{L_n}(u) d\sigma(u).$$

The question reproduced below asks whether it is compatible with $\mathbf{ZF}$ that separate additivity, rotation invariance, and vanishing whenever two arguments meet only at the origin characterize the scalar multiples of $\tilde{V}$.

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CARLOS H. JIMÉNEZ AND IGNACIO VILLANUEVA

The result above includes and extends the main results in [10]. To see this, note that, as the authors mention in that paper, additive functions taking values in $[0, \infty)$ are always positively homogenous.

In that same paper, the authors study the following question

QUESTION 3.5. *Let $F : \mathcal{S}_0^n \rightarrow \mathbb{R}$ be separately additive, rotation invariant, and vanishes when the intersection of two of the arguments is $\{o\}$. Does there exist a constant $c \in \mathbb{R}$ such that $F(L_1, \dots, L_n) = c\tilde{V}(L_1, \dots, L_n)$?*

They show that, thus formulated, in the presence of the Axiom of Choice the question is false. The counterexample follows from the existence of additive non linear functions. Next, they ask if a positive answer to the question is compatible with $\mathbf{ZF}$. A path to answer this could be to study if the reasonings in [20] apply in the Solovay model, since in that case additive functions are automatically linear and continuous.

For an integer $n \geq 2$, let D_n denote the assertion that every map

$$F : (\mathcal{S}_0^n)^n \rightarrow \mathbb{R}$$

having those three properties satisfies $F = c\tilde{V}$ for some $c \in \mathbb{R}$.

Theorem 1 (Full relative-consistency answer). *If ZF is consistent, then so is*

$$\text{ZF} + “D_n \text{ holds for every integer } n \geq 2”.$$

More concretely, every model of

$$\text{ZF} + \text{DC} + “\text{every set of reals has the Baire property}” \quad (*)$$

satisfies D_n simultaneously for all $n \geq 2$.

Idea of the proof

Radial functions identify star bodies with the positive cone of the separable Banach space $C(S^{n-1})$. Extend the given functional additively to the whole space. In an all-sets-Baire-property model, each one-variable section is a Baire-measurable homomorphism between Polish groups, hence continuous by Pettis’ theorem. This supplies positive homogeneity and exactly the separate boundedness missing from the question. The bounded characterization already proved in the source paper then gives integration against a measure on the sphere, and rotation invariance makes that measure spherical Lebesgue measure up to scale.

The proof separates the set-theoretic input from the functional-analytic argument. The key point is that the sections of F live not merely on $\mathbb{R}$, but on the infinite-dimensional Polish group $C(S^{n-1})$.

2 Automatic continuity

We use two standard consequences of (*).

Lemma 1 (Baire transfer and Pettis continuity). *Assume (*). Then every subset of any Polish space has the Baire property. Consequently every additive group homomorphism*

$$h : X \longrightarrow \mathbb{R}$$

from a separable real Banach space X is continuous and hence $\mathbb{R}$ -linear.

Proof. The first statement is the standard Baire-isomorphism theorem: modulo meagre sets, every uncountable Polish space is Baire-isomorphic to Baire space ω^ω , which is homeomorphic to the irrationals. Countable Polish spaces cause no problem. Thus the hypothesis for subsets of the reals transfers to every Polish space. The usual construction of this isomorphism is countable, so DC suffices.

For completeness, let $h : X \rightarrow \mathbb{R}$ be additive and put

$$A_k = h^{-1}([k, k+1)), \quad k \in \mathbb{Z}.$$

These Baire-property sets cover the Baire space X , so some A_k is nonmeagre. Pettis’ category lemma gives a neighbourhood U of zero contained in $A_k - A_k$. Hence $|h(x)| \leq 1$ on U . For any $m \geq 1$, $m^{-1}U$ is a neighbourhood of zero and additivity gives $|h(x)| \leq 1/m$ there. Thus h is continuous at zero. It is rationally homogeneous by additivity; approximation of a real scalar by rationals and continuity make it $\mathbb{R}$ -linear. $\square$

The invocation of Pettis is harmless in $\text{ZF} + \text{DC}$: its proof is the standard countable Baire-category argument. The same observation supplies the Baire category theorem and uniform boundedness for separable Banach spaces.

3 Proof of the characterization

Proof of Theorem 1 inside ().* Fix $n \geq 2$ and a functional F as in D_n . Radial functions identify S_0^n with the positive cone $C(S^{n-1})_+$, and radial addition becomes pointwise addition.

Separate additivity extends F uniquely, coordinate by coordinate, to a separately additive map

$$T : C(S^{n-1})^n \longrightarrow \mathbb{R}. \quad (1)$$

This is just the Grothendieck-group construction. Explicitly, write $f = f^+ - f^-$ and expand in each coordinate with alternating signs. The result does not depend on a chosen decomposition: if $u - v = u' - v'$ in $C(S^{n-1})_+$, then $u + v' = u' + v$, and cancellation plus additivity gives the same value. This construction uses no choice principle.

Fix all but one argument in (1). The remaining section $h : C(S^{n-1}) \rightarrow \mathbb{R}$ is an additive homomorphism between Polish groups. Lemma 1 makes it continuous and $\mathbb{R}$ -linear. This holds in every coordinate and for every choice of the other arguments. Restricting back to $C(S^{n-1})_+$ yields two facts that were absent from the question's hypotheses:

1. F is positively homogeneous in every coordinate;
2. F is separately bounded on star bodies contained in the Euclidean unit ball.

Indeed, after the other arguments are fixed, the corresponding continuous linear functional on $C(S^{n-1})$ is bounded on its unit ball.

We may now apply the separately bounded branch of Theorem 3.4 of Jiménez–Villanueva. That theorem states precisely that a separately additive, positively homogeneous, separately bounded F which vanishes when two arguments intersect only at the origin has a representation

$$F(L_1, \dots, L_n) = \int_{S^{n-1}} \rho_{L_1}(u) \cdots \rho_{L_n}(u) d\mu(u) \quad (2)$$

for a finite signed regular Borel measure μ . Rotation invariance of F makes μ rotation invariant. Uniqueness of normalized invariant measure on the compact homogeneous sphere gives $\mu = a\sigma$ for some $a \in \mathbb{R}$. Comparison with the normalization of $\tilde{V}$ yields

$$F(L_1, \dots, L_n) = na \tilde{V}(L_1, \dots, L_n),$$

which is D_n .

There is no hidden use of full Choice in this last paragraph. For the bounded case, the source proof uses joint continuity of a finite multilinear form, the Riesz representation theorem on the compact metric sphere, countable regularity, and the continuous orthosymmetric $C(K)$ representation lemma. These are all available under DC: joint continuity follows from the Baire-category/uniform-boundedness argument, and the measure constructions are countable on a compact metrizable space. Equivalently, (2) follows by iterating the standard identity $B(f, g) = B(fg, 1)$ for continuous orthosymmetric bilinear forms on $C(K)$ and then applying compact-metric Riesz representation. $\square$

4 Consistency strength and scope

Solovay's classical model would already prove compatibility, assuming an inaccessible cardinal. The Baire-property input is substantially cheaper. Shelah proves in Conclusion 7.17 of *Can you take Solovay's inaccessible away?* that the following theories are equiconsistent:

$$\text{ZFC} \quad \text{and} \quad \text{ZF} + \text{DC} + \text{“every set of reals has the Baire property”}. \quad (3)$$

Moreover ZF and ZFC are equiconsistent, by passing to the constructible inner model. Thus (3) and the internal proof establish Theorem 1 from Con(ZF).

Remark 1 (Exact logical scope). *This proves a full affirmative answer to the compatibility question. It does not prove D_n in bare ZF, nor does it prove independence. The Axiom of Choice gives the known counterexample $F = h \circ \tilde{V}$ from a nonlinear Cauchy additive $h : \mathbb{R} \rightarrow \mathbb{R}$, whereas the Baire-property model eliminates not only such h but every discontinuous additive section on $C(S^{n-1})$.*

Remark 2 (Why scalar Cauchy regularity alone is not the proof). *Knowing only that additive maps $\mathbb{R} \rightarrow \mathbb{R}$ are linear controls scalar rays but does not itself bound a functional on the infinite-dimensional space $C(S^{n-1})$. The all-sets-Baire-property hypothesis is what makes every section $C(S^{n-1}) \rightarrow \mathbb{R}$ continuous, exactly supplying the source theorem’s missing hypothesis.*

Novelty check

Bounded exact-title, exact-question, phrase, author, citation, and Solovay-model searches on 11 August 2026 found the two source papers and later work on other dual-volume questions, but no later resolution of Question 3.5. The argument is short enough that it may be folklore; novelty confidence is therefore moderate pending expert review.

References

- [1] C. H. Jiménez and I. Villanueva, *Characterization of dual mixed volumes via polymeasures*, J. Math. Anal. Appl. 426 (2015), 688–699; [arXiv:1408.6796](#).
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- [4] R. M. Solovay, *A model of set-theory in which every set of reals is Lebesgue measurable*, Ann. of Math. 92 (1970), 1–56.